4. A bank pays 8% per annum interest compounded quarterly. Find the equal deposits to be made at the end of each quarter for 10 years to have ₹30,200? $[(1.02)^{40} = 2.2080]$.

5. A person deposits ₹2,000 at the end of every month from his salary towards his contributory pension scheme. The same amount is credited by his employer also. If 8% rate of compound interest is paid, then find the maturity amount at end of 20 years of service.
   $[(1.0067)^{240} = 4.9661]$.

6. Find the present value of ₹2,000 per annum for 14 years at the rate of interest of 10% per annum. If the payments are made at the end of each payment period. $[(1.1)^{-14} = 0.2632]$.

7. Find the present value of an annuity of ₹900 payable at the end of 6th month for 6 years. The money compounded at 8% per annum. $[(1.04)^{-12} = 0.6252]$.

8. Find the amount at the end of 12 years of an annuity of ₹5,000 payable at the beginning of each year, if the money is compounded at 10% per annum. $[(1.1)^{12} = 3.1384]$.

9. What is the present value of an annuity due of ₹1,500 for 16 years at 8% per annum? $[(1.08)^{-16} = 0.2919]$.

10. What is the amount of perpetual annuity of ₹50 at 5% compound interest per year?

7.2 Stocks, Shares, Debentures and Brokerage

To start any big business, a large sum of money is needed. It is generally not possible for an individual to manage such a large sum. Therefore the total sum of money can be divided into equal parts called **shares**. The holder of shares are called **shareholders**.

7.2.1 Types of shares:

There are two type shares namely **common** (or equity) and **preferred**.

Generally, the profit gained by the company is distributed among their share holders The preferred share holders have a first claim on dividend. When they have been paid, the remaining profit is distributed among the common share holders.

7.2.2 Definitions

(i) **Capital stock** is the total amount invested to start a company.

(ii) The share purchased by the individual is also called **stock**.

(iii) The persons who buy the shares are also called **stock holders**

(iv) **Face value :** The original value of a share at which the company sells it to investors is called a face value or nominal value or par value. It is to be noted that the original value of share is printed on the share certificate.

(v) Market value : The price at which the stock is bought or sold in the market is called the market value (or cash value).



The market value of a share keeps on changing from time to time.

07_11th_BM-STAT_Ch-7-EM.indd 167 21-04-2020 12:24:23 PM